

👉 XAI: Practitioner Interpretability Needs

FAMAF - UNC

November 3, 2025



**eXplainable
Artificial Intelligence**

Stage 3: Large Scale Survey ---

Objective: Observe whether data scientists can use interpretability tools to **uncover** the injected issues

Study Design

- **Type:** Survey based on example queries from previous tools
- **Participants:** 197 from mailing list of large tech company
- **Analysis:**
 - ▶ Coded open-ended responses
 - ▶ Statistical tests to compare outcomes by condition

🍷 Stage 3: Large Scale Survey - Experimental Conditions ---

Explanation Type

- GAM
- SHAP

Visualization Type

- Normal (correct)
- Manipulated (obviously wrong)

Key Question

Do people trust obviously wrong explanations less?



🍷 Stage 3: Large Scale Survey - # Result 1: ---

Performance with Explanations

Key Findings:

- **GAM » SHAP**
 - ▶ GAM users performed significantly better
- **Better results with good explanations than manipulated**
 - ▶ But effect was smaller than expected

Implication

People don't always detect obviously flawed explanations

🍷 Stage 3: Large Scale Survey - Result 2: ---

How Practitioners Make Deployment Decisions

① Intuition-Based Decisions

- ▶ Made decisions based on gut feeling

② Superficial Justification

- ▶ Used explanations to justify pre-existing beliefs

③ Critical Examination (Some)

- ▶ Small group used tools as intended

Design Challenge

How to push people towards deliberative reasoning?

🍷 Stage 3: Large Scale Survey - # Result 3: ---

Mental Models of Tools Understanding vs. Confidence

- Participants largely **did not understand tools well**
- Despite that, they **believed tools effective** for many uses

Critical Question

Is it bad for explanations to persuade people without understanding?

🍲 Stage 3: Large Scale Survey - Result 4: ---

Experience Paradox The Experience-Confidence Trade-off

- More ML background → Better understanding of explanations
- More ML experience → Less confidence in explanations
- Less confidence → Lower willingness to deploy

Challenge

How do we make ML explanations more accessible without sacrificing quality?

Key Takeaways from Both Papers

Paper 1: What Users Need

- Cooperative interpretability
- Process-oriented tools
- Context-dependent explanations
- Post-deployment support

Paper 2: How Tools Perform

- Over-trust is common
- Misuse is frequent
- Better tools needed
- Training is essential

Critical Questions for Discussion

- ① **Better tools vs. better training?**
 - ▶ Should we focus on improving tools or educating users?
- ② **Persuasion without understanding?**
 - ▶ Is it problematic if explanations convince without comprehension?
- ③ **Accessibility vs. sophistication?**
 - ▶ How to make tools accessible without oversimplifying?
- ④ **Role-specific design?**
 - ▶ Should we design different tools for different roles?

Implications for Interpretability Research

Research Directions:

- Design tools with specific user roles and stages in mind
- Account for social and organizational context
- Provide safeguards against misuse
- Balance complexity with accessibility
- Support continuous engagement throughout ML lifecycle

Open Research Questions ---

- ➊ How can we design interpretability tools that are both powerful and accessible?
- ➋ What interventions can reduce over-trust in explanations?
- ➌ How should interpretability tools adapt to different user expertise levels?
- ➍ What role should organizational culture play in interpretability tool design?
- ➎ How can we better evaluate whether tools actually help users?

Conclusion ---

Main Messages:

- Interpretability tools are widely used but often misused
- Gap exists between what researchers build and what practitioners need
- Context, role, and stage matter significantly
- User training is as important as tool design
- Much work remains to bridge research and practice

Thank You!

Questions and Discussion